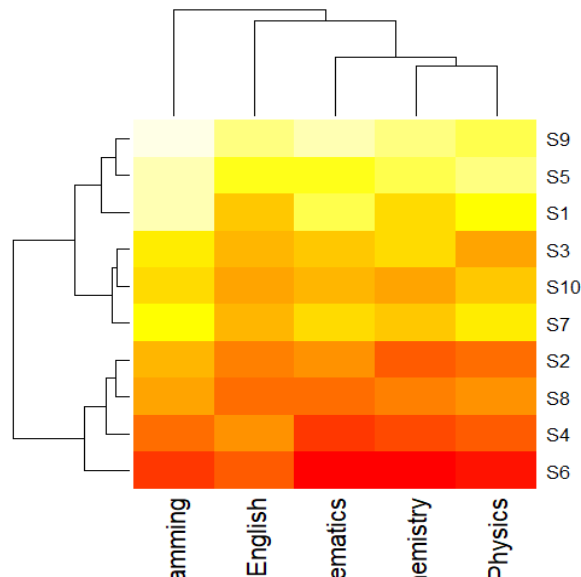


Activity 1: Heatmap – Student Performance Analysis

1. Construct a Heatmap

```
student <- data.frame(  
+   Mathematics = c(92,76,81,65,90,58,84,71,95,79),  
+   Physics = c(88,72,78,70,94,62,86,75,91,81),  
+   Chemistry = c(84,69,83,68,91,60,82,73,93,77),  
+   Programming = c(95,80,85,72,96,65,88,77,98,83),  
+   English = c(81,74,79,75,89,70,80,72,94,78)  
+ )
```



2. Identify highest overall performer.

```
rownames(student) <- paste("S",1:10,sep="")  
> student  
  Mathematics Physics Chemistry  
S1           92      88        84  
S2           76      72        69  
S3           81      78        83  
S4           65      70        68  
S5           90      94        91  
S6           58      62        60  
S7           84      86        82  
S8           71      75        73  
S9           95      91        93  
S10          79      81        77  
 Programming English  
S1           95      81  
S2           80      74  
S3           85      79  
S4           72      75  
S5           96      89  
S6           65      70  
S7           88      80  
S8           77      72  
S9           98      94  
S10          83      78  
> heatmap(as.matrix(student),  
+         col = heat.colors(20),  
+         scale = "none")  
> student$Average <- rowMeans(student)
```

3. Find highest average subject.

```
colMeans(student[,1:5])
```

Mathematics 79.1

Physics 79.7

Chemistry 78.7

Programming 83.9

English 79.2

4. Identify students needing support.

```
student[student$Average < 75, ]
```

Students needing support:

- S4
- S6
- S8

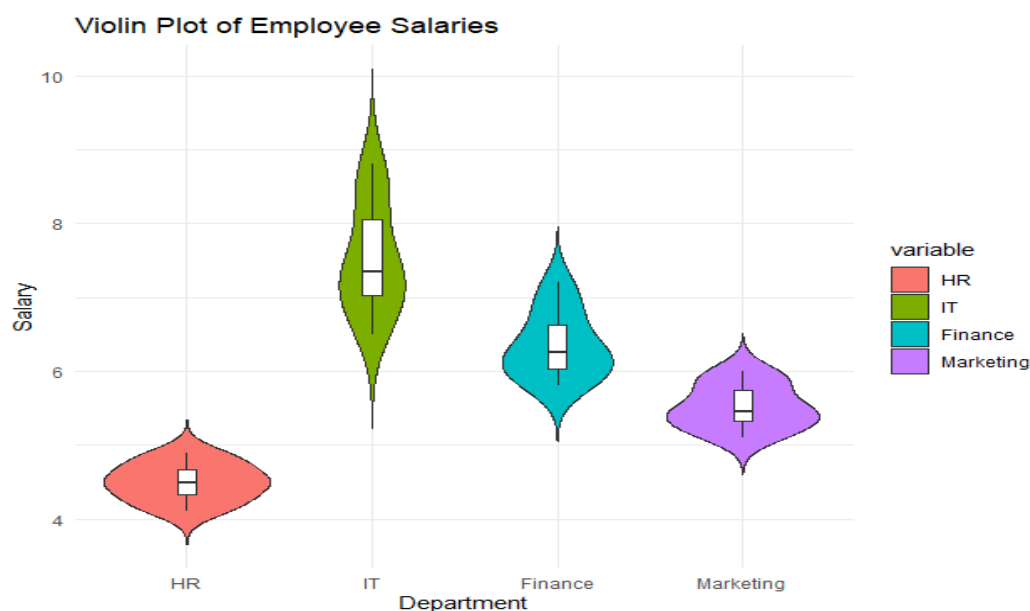
5. Explain Heatmap advantages.

A heatmap is useful because:

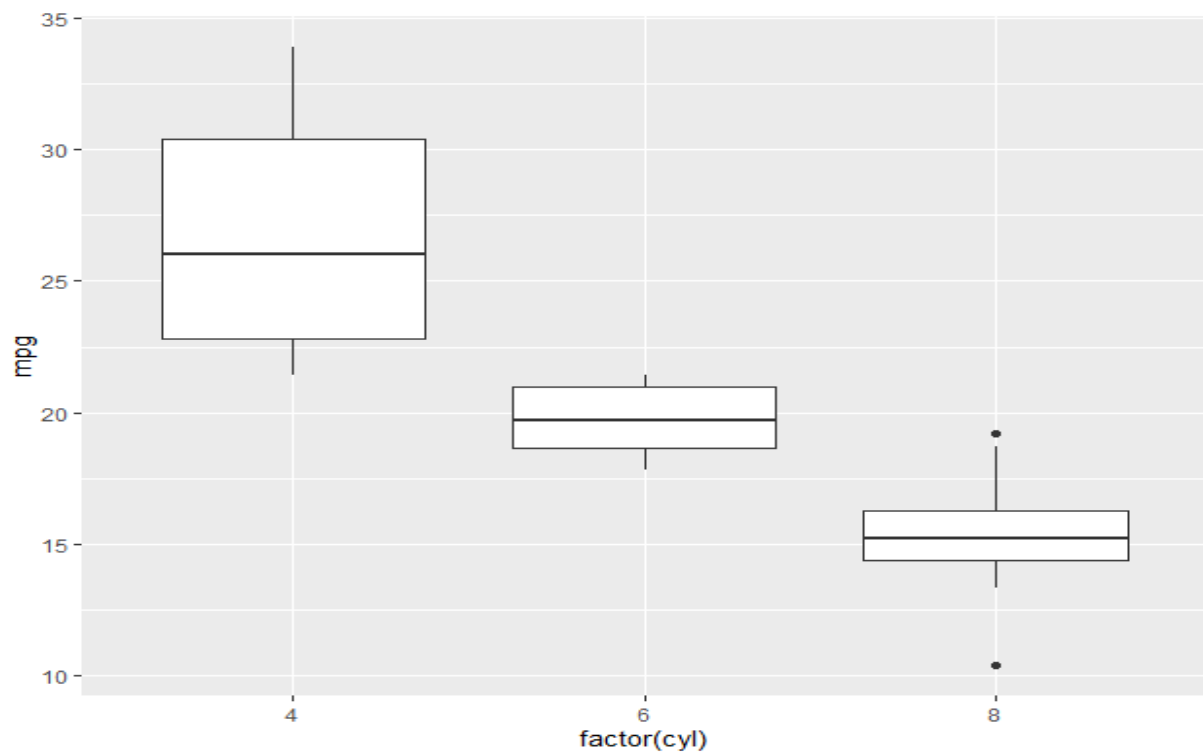
1. It represents large amounts of data visually.
2. It uses colors to quickly identify high and low performance.
3. It makes comparison between students and subjects easier.
4. It helps detect patterns and weak areas.
5. It supports quick decision-making.

Activity 2: Violin Plot & Boxplot – Employee Salary Analysis

1. Draw violin plots.



2. Construct boxplots.



3. Largest variation?

```
apply(salary[,2:5], 2, sd)
```

HR 0.26

IT 0.74

Finance 0.47

Marketing 0.31

4. Highest median?

```
apply(salary[,2:5], 2, median)
```

HR 4.5

IT 7.35

Finance 6.25

Marketing 5.45

5. Which plot better represents distribution?

Violin Plot

Reason:

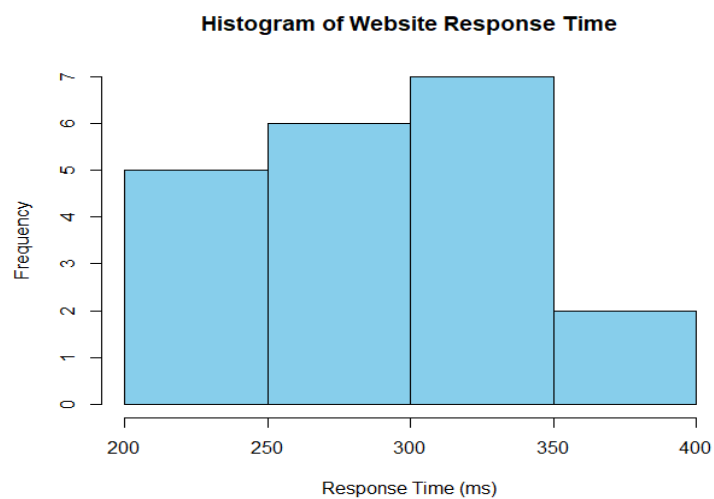
- Shows the full distribution (density) of the data.
- Displays peaks and spread more clearly.
- Also indicates the median and quartiles when combined with a boxplot.

A **boxplot** is better for identifying:

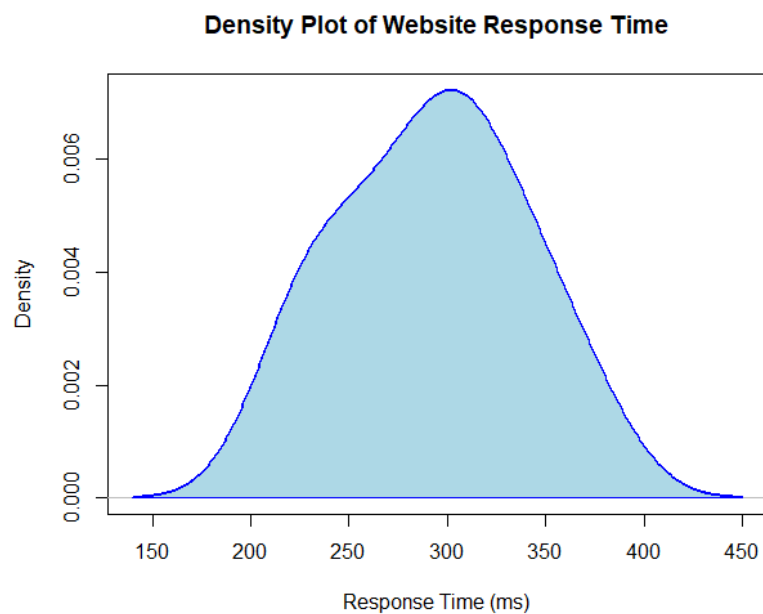
- Median
- Quartiles
- Outliers

Activity 3: Histogram & Density Plot – Website Response Time

1. Draw histogram.



2. Plot density.



3. Determine skewness.

```
library(moments)
> skewness(response)
[1] -0.007827051
```

4. Common range.

Most of the observations lie between:

250 and 330 ms

5. Which plot better represents distribution?

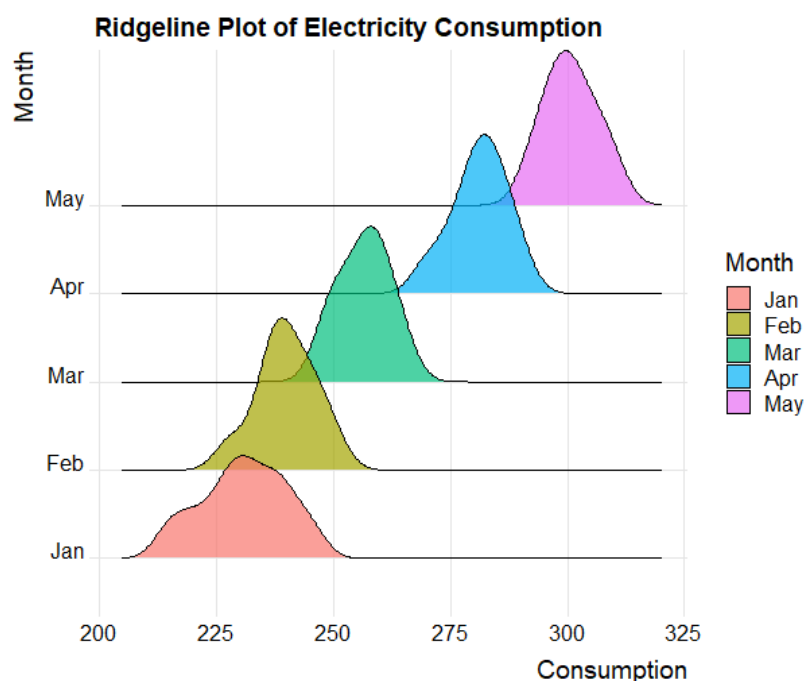
Density Plot

Reason:

- Shows the overall shape of the distribution smoothly.
- Makes it easier to identify peaks and skewness.
- Better for understanding the distribution than a histogram when the dataset is small.

Activity 4: Ridgeline Plot – Electricity Consumption

1. Create ridgeline plot.



2. Highest average month?

```
colMeans(electricity[,2:6])
Jan Feb Mar Apr May
230.8 239.9 256.5 281.2 300.7
```

3. Widest spread?

```
apply(electricity[,2:6],2,sd)
```

Jan	Feb	Mar
9.223159	6.279597	5.380004
Apr	May	
5.865151	5.538752	

4. Describe trend.

Electricity consumption shows a continuous increasing trend from January to May.

- January → Lowest consumption
- February → Slight increase
- March → Moderate increase
- April → Large increase
- May → Highest consumption

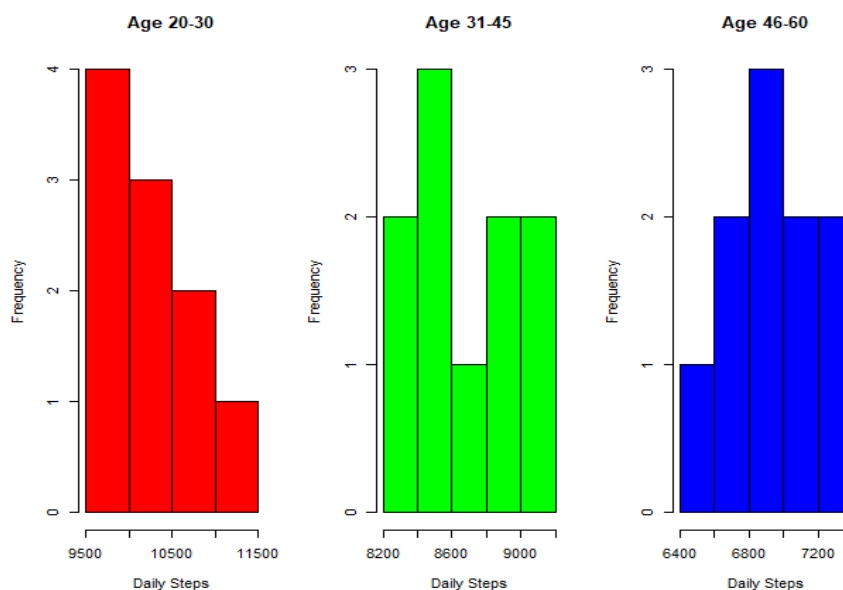
5. Why ridgeline?

A ridgeline plot is useful because:

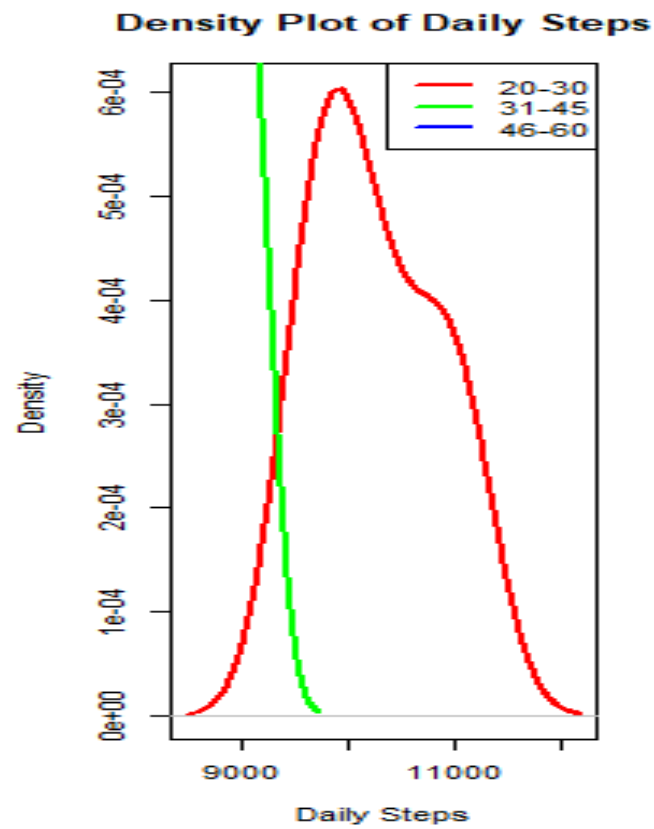
- It compares distributions across multiple groups.
- It shows changes over time clearly.
- It helps identify shifts, peaks, and spread.
- It allows easy comparison between months.

Activity 5: Histogram, Density & Boxplot – Daily Steps

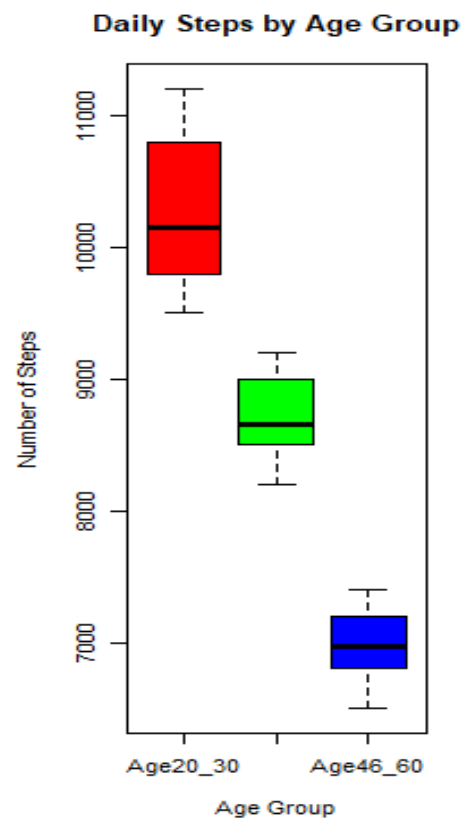
1. Draw histograms



2. Plot density curves.



3. Create boxplots.



4. Highest median group?

```
apply(steps[,2:4], 2, median)
Age20_30 Age31_45 Age46_60
10150      8650      6975
```

5. Best visualization and justify.

Boxplot

Reason:

- Clearly compares the three age groups.
- Shows median values easily.
- Displays spread and variation.
- Helps identify possible outliers.

A density plot shows the distribution shape, and histograms show frequency, but the **boxplot** is best for comparing multiple groups.